Architectural Foundations of Distributed Consensus and Fault-Tolerant Storage
Systems Engineering Technical Report
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1. Overview of Consensus Protocols
In distributed state machine replication, ensuring safety and liveness across asynchronous networks subject to network partitions is governed by the CAP theorem and the FLP impossibility result. While Multi-Paxos has served as the theoretical gold standard, its operational complexity and underspecified cluster membership changes led to the widespread adoption of the Raft consensus algorithm.

Raft decomposes distributed consensus into three distinct subproblems: leader election, log replication, and safety invariants. Raft guarantees that at any given term, at most one leader can be elected, and leader election requires an affirmative vote from a strict majority quorum: Q = floor(N/2) + 1.

2. Election Timeouts and Heartbeat Mechanisms
To prevent split-vote scenarios, Raft utilizes randomized election timeouts between 150ms and 300ms. The leader continuously broadcasts periodic heartbeats with an interval of 50ms to maintain authority. If a follower receives no heartbeat within its randomized election timeout window, it transitions to the Candidate state, increments the current term, votes for itself, and broadcasts RequestVote RPCs.

Log compaction is implemented via asynchronous snapshotting. Once the write-ahead log exceeds 50 megabyte the state machine serializes an immutable memory snapshot to persistent NVMe storage and truncates precedin